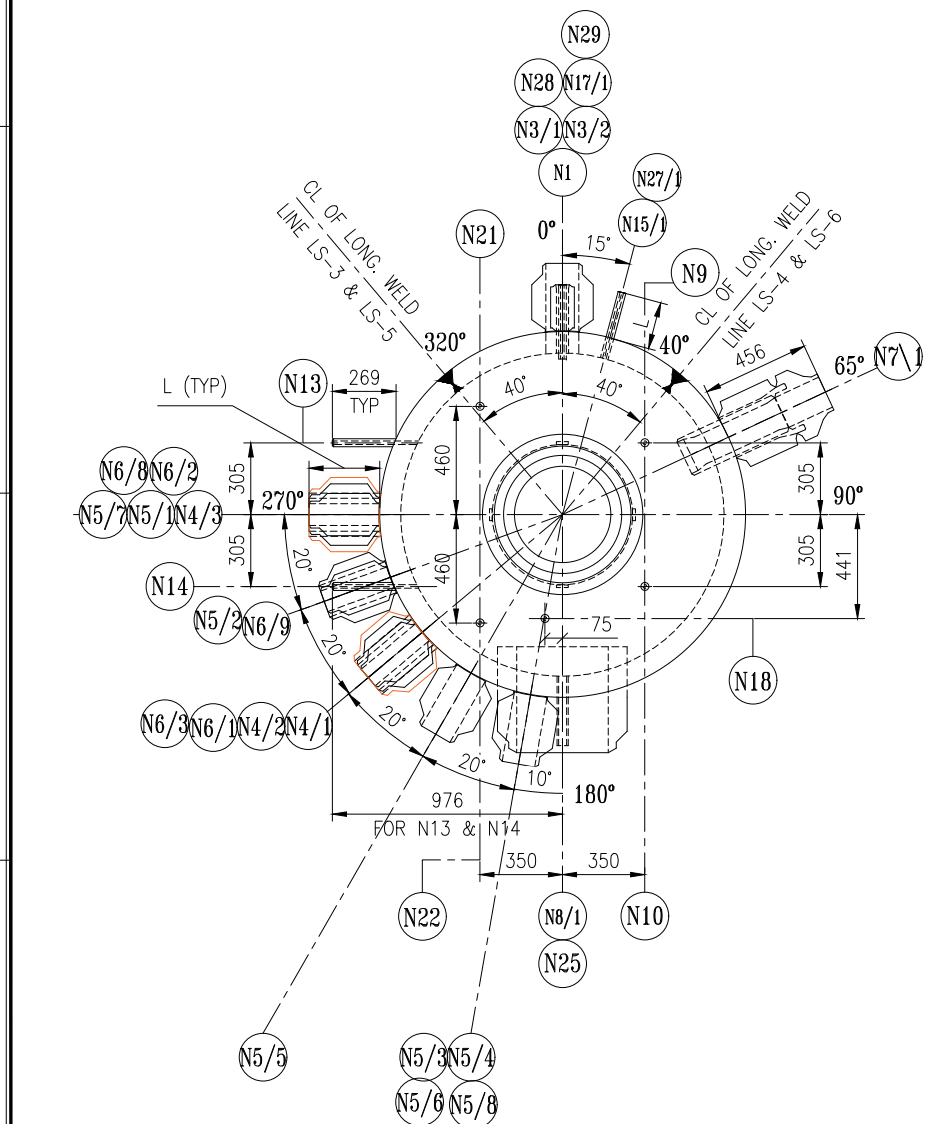
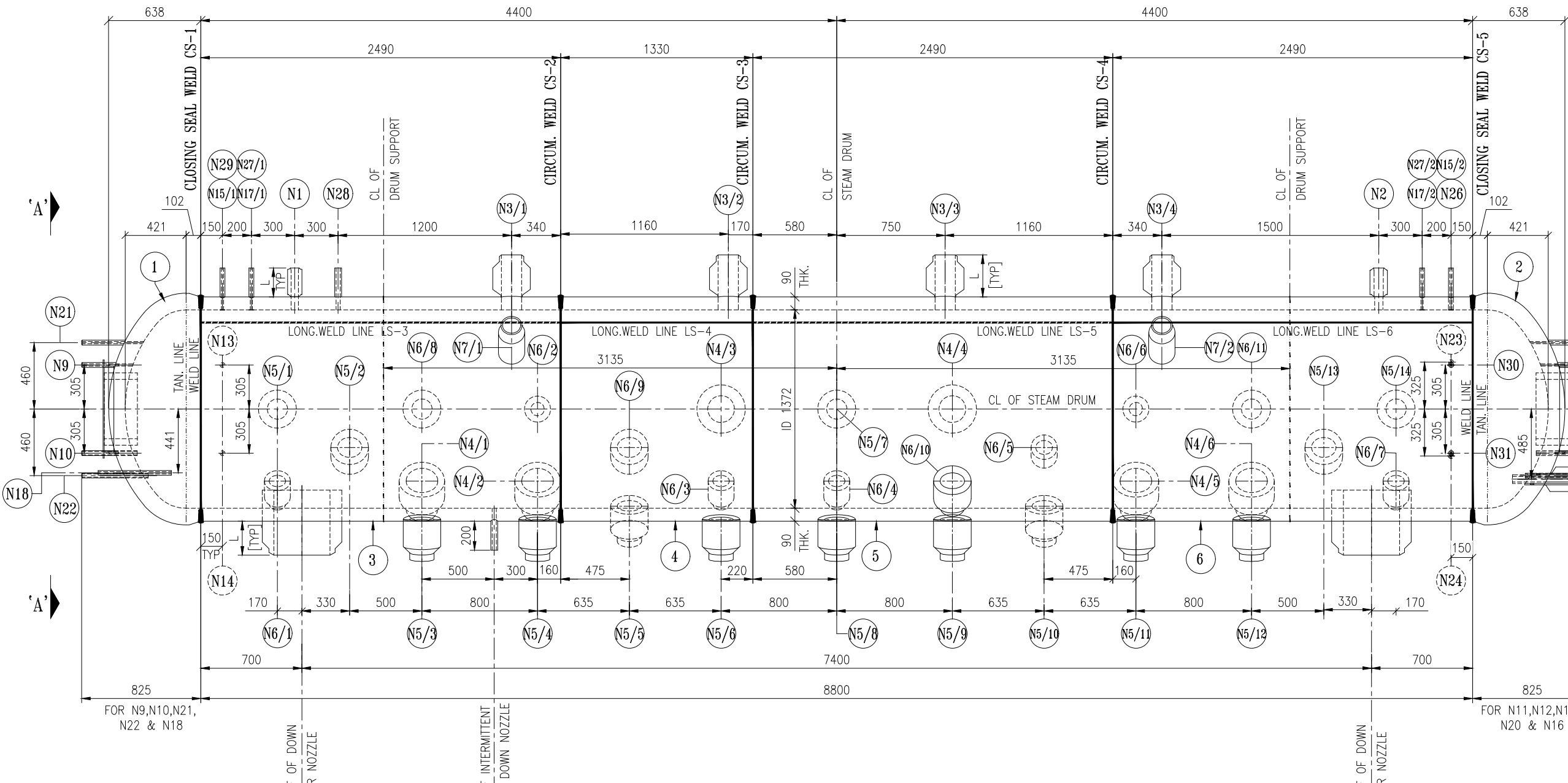


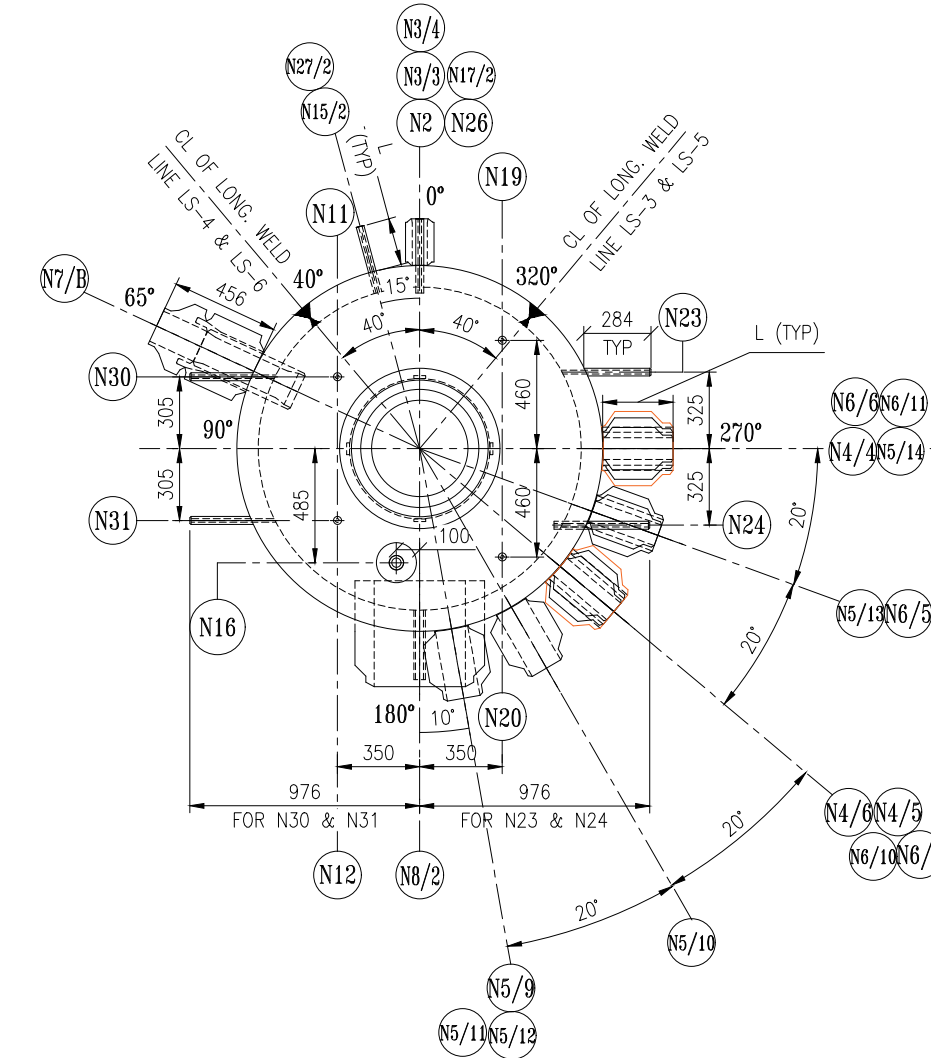
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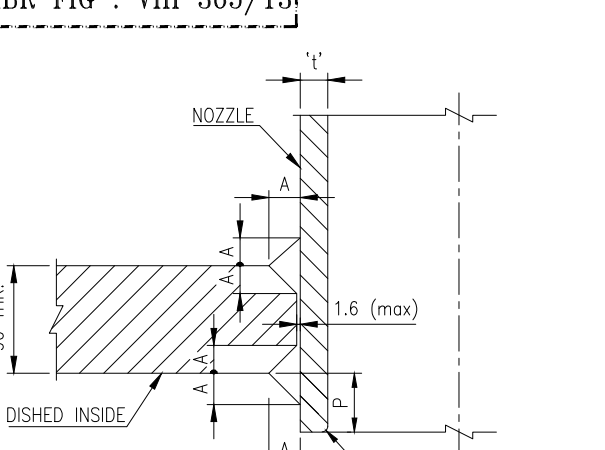
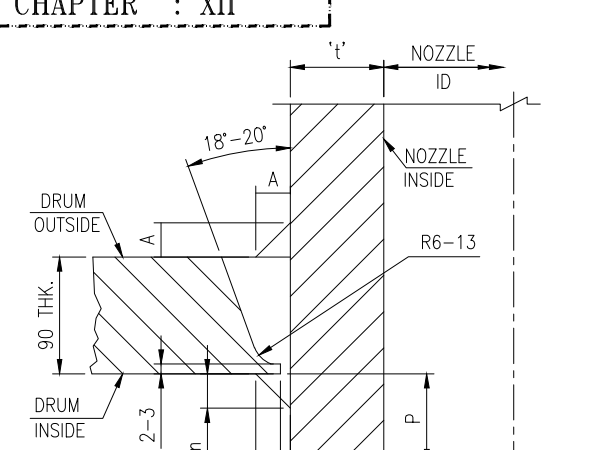
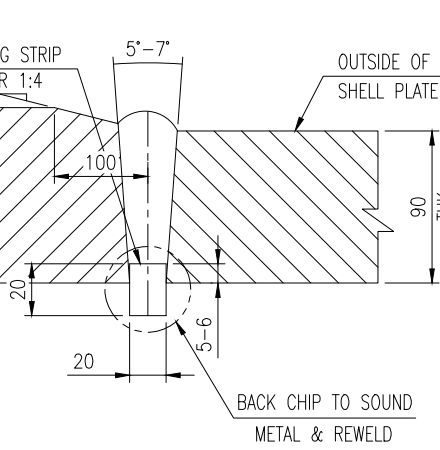
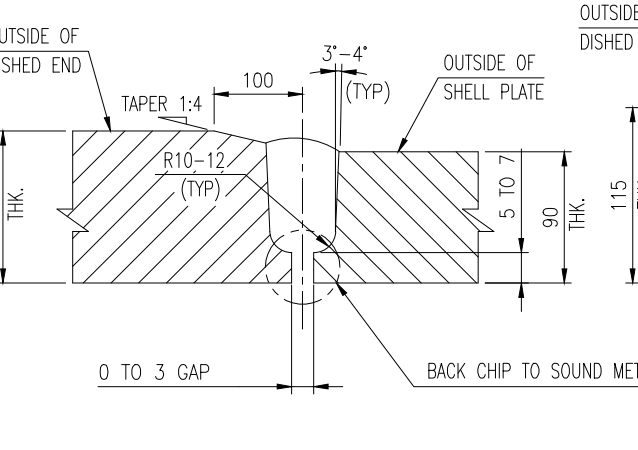
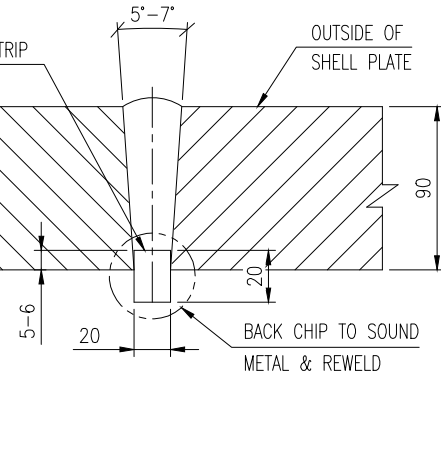
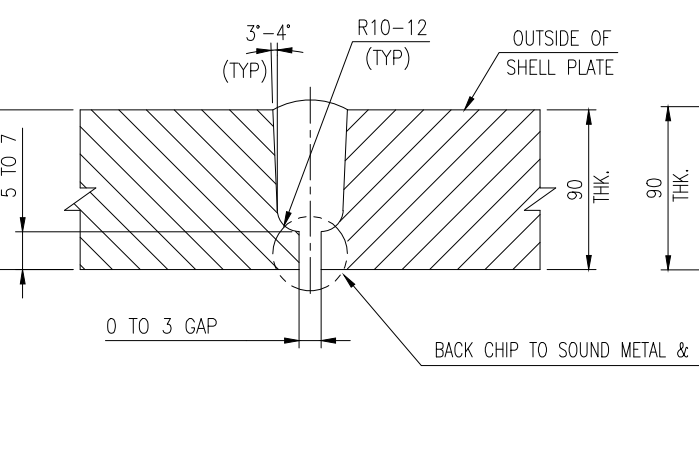
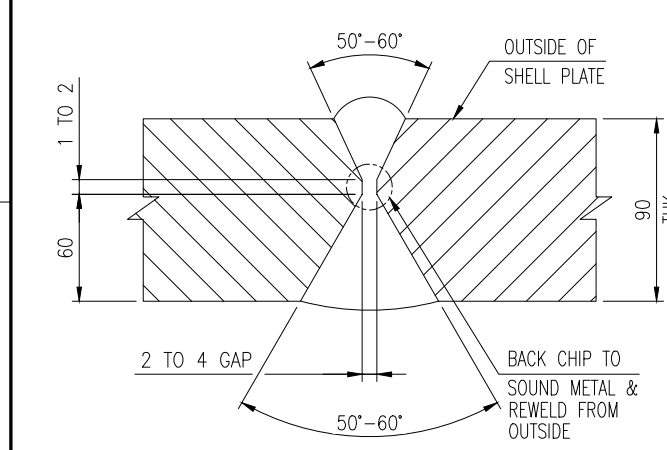
VIEW - 'A A'



ELEVATION
ARRANGEMENT OF STEAM DRUM
[VIEW FROM FRONT OF BOILER]



VIEW - 'B B'



WELD PREPARATION OF LONGITUDINAL SEAM
FOR STEAM DRUM
(SEAM MK. Nos. LS-3 TO LS-6)

WELD PREPARATION OF CIRCUMFERENTIAL SEAM
(SEAM MK. Nos. CS-2 TO CS-4)

ALTERNATE DETAIL

WELD PREPARATION OF CLOSING SEAM
FOR STEAM DRUM MARK Nos. CS-1 & CS-5

ALTERNATE DETAIL

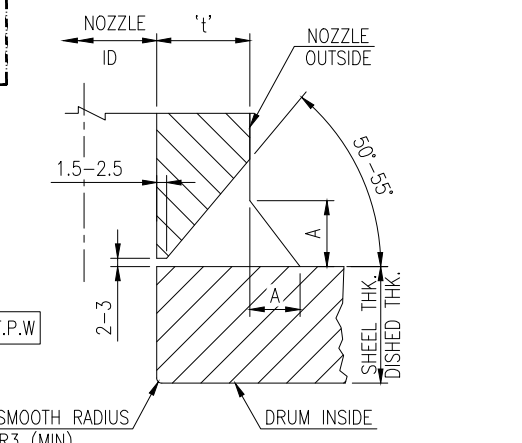
WELDING DETAIL OF NOZZLE TO DRUM
(FOR MK.No. N8/1 & N8/2)

WELDING DETAIL OF NOZZLE TO DRUM
[FOR MK.No. N18]

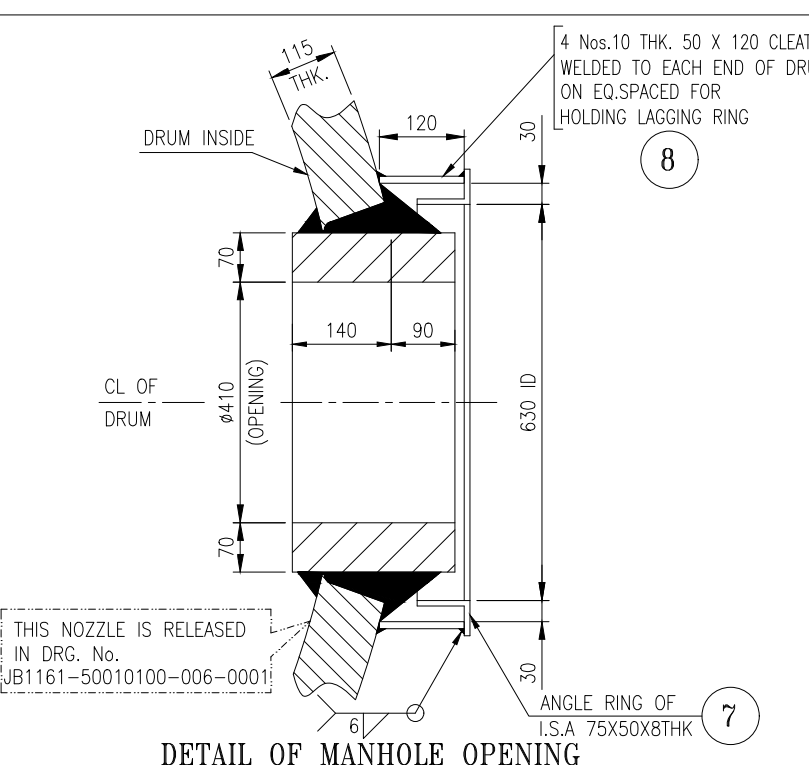
TABLE - 1

Shos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (mm)	Nozzle Thickness (mm)	Connecting Pipe Thickness (mm)	Projection from End of Drum	Drum Vertical Axis (mm)	Drum Horizontal Axis (mm)	Nozzle Length (mm)	Nozzle Radius (mm)	M/C Base for End Preparation (mm)	Projection from End of Nozzle (mm)
1	N1	SAFETY VALVE NOZZLE - 2	STEAM DRUM SHELL	1	140	60	14.27	300	200	200	44.80	0	200	38.00
2	N2	SAFETY VALVE NOZZLE - 2	STEAM DRUM SHELL	1	140	60	14.27	300	200	200	44.80	0	200	38.00
3	N3/1 TO N3/4	SATURATED STEAM STUB	STEAM DRUM SHELL	4	260	60	14.27	300	330	300	140.00	60	0	0
4	N4/1 TO N4/6	COMBUSTOR SIDE WALL RISER STUB	STEAM DRUM SHELL	6	315	66	18.26	300	330	300	183.00	66	0	0
5	N5/1 TO N5/14	COMBUSTOR FRONT WALL RISER STUB	STEAM DRUM SHELL	14	260	60	14.27	300	330	300	140.00	60	0	0
6	N6/1 TO N6/7	SEPARATOR SIDE, REAR & PARTITION WALL RISER STUB	STEAM DRUM SHELL	7	180	44	11.13	300	330	300	92.04	44	0	0
7	N6/8 TO N6/11	SEPARATOR FRONT WALL RISER STUB	STEAM DRUM SHELL	4	260	60	14.27	300	330	300	139.76	60	0	0
8	N7/1 & N7/2	FEED INLET NOZZLE WITH THERMAL SLEEVE	STEAM DRUM SHELL	2	550	81	34.93	250	515	465	387.34	81	125	125
9	N8/1 & N8/2	DOWNCOMER NOZZLE	STEAM DRUM SHELL	2	550	81	34.93	250	515	465	387.34	81	125	125
10	N9	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
11	N10	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
12	N11	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
13	N12	LEVEL TRANSMITTER (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
14	N13	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
15	N14	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	350	305	337	24.30	10	0	0
16	N15/1 & N15/2	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
17	N16	CHEMICAL INJECTION NOZZLE WITH THERMAL SLEEVE	STEAM DRUM DISHED END	1	170	46.4	REF. DRG. 009-0001	100	485	REF. DRG. No.009-0001	100	100	100	100
18	N17/1	PRESSURE GAUGE NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
19	N17/2	REMOTE PRESSURE GAUGE NOZZLE AT FLOOR	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
20	N18	CBD NOZZLE	STEAM DRUM DISHED END	1	33.4	9.09	4.55	75	441	659	629	24.30	10	200
21	N19	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	460	407	377	24.30	10	0
22	N20	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	460	407	377	24.30	10	0
23	N21	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	460	407	377	24.30	10	0
24	N22	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED END	1	33.4	9.09	4.55	350	460	407	377	24.30	10	0
25	N23	HYDRA STEP NOZZLE (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	325	310	280	24.30	10	0	0
26	N24	HYDRA STEP NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	325	310	280	24.30	10	0	0
27	N25	INTERMITTENT BLOW DOWN NOZZLE	STEAM DRUM SHELL	1	48.3	10.15	5.08	200	230	200	38.14	10	0	0
28	N26	NITROGEN BLANKETING	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
29	N27/1 & N27/2	AIR VENT NOZZLE	STEAM DRUM SHELL	2	48.3	10.15	5.08	200	230	200	38.14	10	0	0
30	N28	SPARE STUB	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
31	N29	ITC NOZZLE	STEAM DRUM SHELL	1	33.4	9.09	4.55	200	230	200	24.30	10	0	0
32	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	9.09	4.55	305	300	270	24.30	10	0	0
33	N31	LEVEL TRANSMITTER (WATER)	STEAM DRUM SHELL	1	33.4	9.09	4.55	305	300	270	24.30	10	0	0

IBR CHAPTER : XII
IBR FIG : XII/85E B2

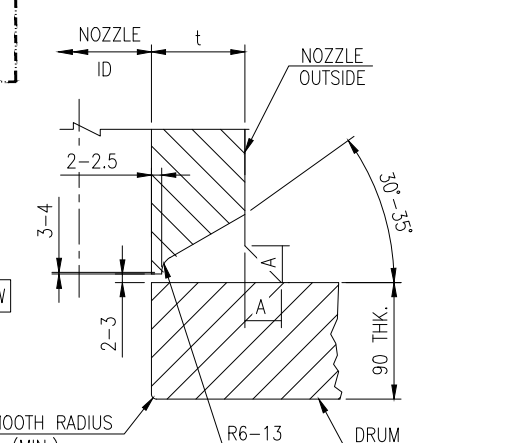


WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK Nos. N9,N10, N11, N12, N13, N14, N15/1, N15/2, N17/1, N17/2, N19, N20, N21, N22, N23, N24, N25, N26, N27/1, N27/2, N28, N29, N30 & N31]

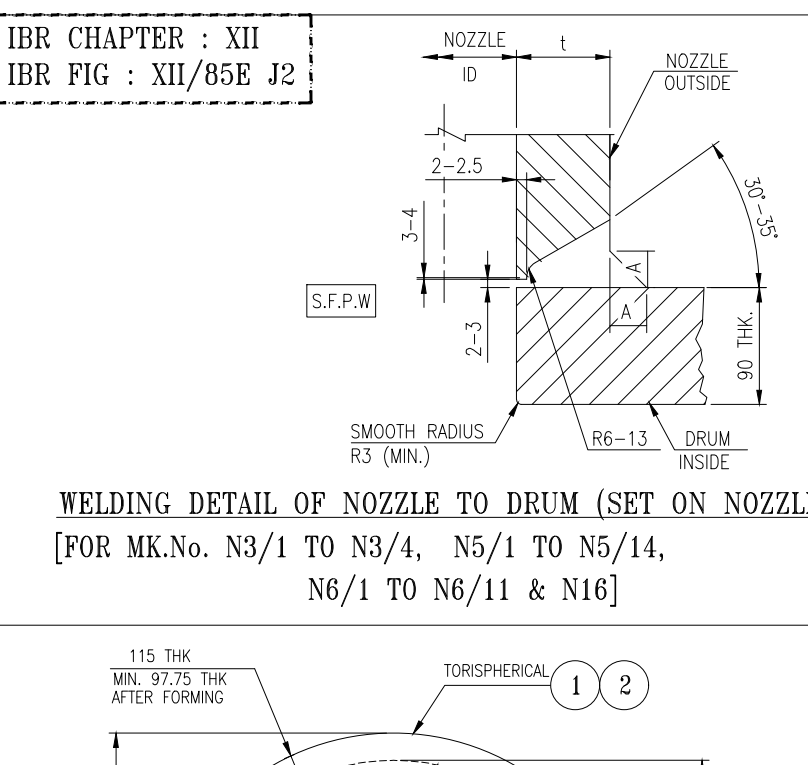


DETAIL OF MANHOLE OPENING

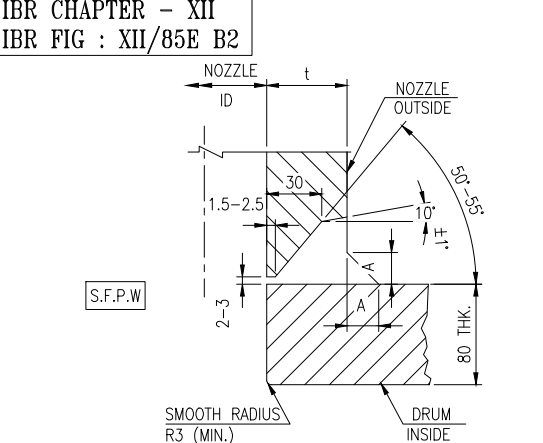
IBR CHAPTER : XII
IBR FIG : XII/85E J2



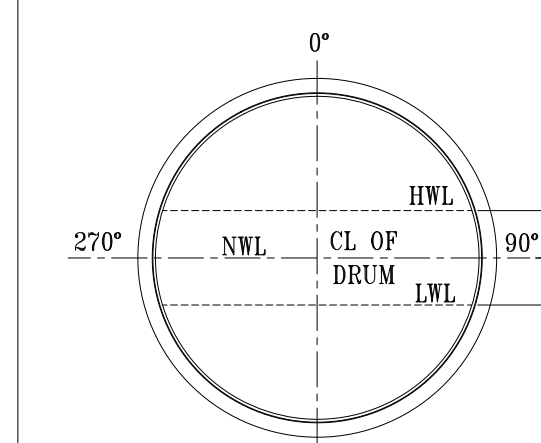
WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MARK Nos. N1 & N2]



WELDING DETAIL OF NOZZLE TO DRUM (SET ON NOZZLE)
[FOR MK.No. N3/1 TO N3/4, N5/1 TO N5/14, N6/1 TO N6/11 & N16]



WELDING DETAIL OF NOZZLE TO DRUM
[FOR MARK No. N7/1, N7/2 & N4/1 to N4/6]



WATER LEVEL FOR STEAM DRUM

23. DRUM TO BE STAMPED AS FOLLOWS :-
ITEM CODE-SDM001-STEAM DRUM
ISGEC HEAVY ENGINEERING LTD
WORKS No.-PD- 0333 (S)/I.W.T.NO.-6662
YEAR OF MAKE-
TESTED TO-229.5 Kg/Sq. cm (g)
TESTED ON-
MAXIMUM ALLOWABLE WORKING PRESSURE-153.00 Kg/Sq. cm (g)
COMPETENT PERSON'S /
INSPECTING AUTHORITY'S OFFICIAL STAMP
FINAL DATE OF EXAMINATION

PAINTING NOTES:-

01. FOR PAINTING NOTES REFER SPECIFICATION NO. JB1161-50579000-PSH-0001.

PWHT CYCLE :- (FOR STEAM DRUM)

A) LOADING TEMP. (MAX.) = 300°C
B) RATE OF HEATING ABOVE 300°C (MAX.) = 55°C/Hr.
C) SOAKING TEMP. = 610°C±10°C
D) SOAKING TIME (MIN.) = 2 Hrs 45 min.
E) RATE OF COOLING ABOVE 300°C (MAX.) = 55°C/Hr.
F) UNLOADING TEMP.(MAX.) = 300°C
G) RATE OF HEATING AFTER UNLOADING = COOLED IN STILL AIR

DESIGN DATA	
CODE OF CONSTRUCTION	IBR 1950 & ITS LATEST AMENDMENTS UP TO OCTOBER 2019 (SEE NOTE : 02)
MAXIMUM WORKING PRESSURE	153.00 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	343 °C
WORKING METAL TEMPERATURE	343 °C (OUTSIDE OF GAS PATH)
HYDRAULIC TEST PRESSURE	229.5 Kg/Sq. cm (g) (AT SHOP & AS PER IBR AT SITE)
RADIOGRAPHY OF LONG. & CIRCUM. SEAM	100%
STRESS RELIEVING	YES (REFER HEAT TREATMENT CYCLE TABLE)
EVAPORATION OF BOILER (M.C.R.)	170 TPH
TOTAL HEATING SURFACE AREA OF BOILER (EXCLUDING FORCED FLOW ELEMENT)	2855 sq.m (APPROX.)
HEATING SURFACE AREA OF FORCE FLOW SECTION (NON STEAMING)	2700 sq.m (APPROX.)
HEATING SURFACE AREA OF DRUM	NIL (OUT SIDE GAS PATH)
STEAM PRESSURE AT SUPER HEATER OUTLET	126 Kg/Sq. cm (g)
SUPER HEATER OUTLET TEMPERATURE	545 ±5 °C
WEIGHT OF DRUM (EMPTY)	39182.93 Kgs.
INSPECTION	CIB (HARYANA) (SEE NOTE : 03)

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg./Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	85.39	FOR STEAM DRUM
DISHED ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	91.97	FOR STEAM DRUM

GENERAL NOTES :-

01. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
02. THE ZERO DATE OF THIS CONTRACT IS 25.10.2019. HENCE IBR AMENDMENTS UP TO OCTOBER 2019 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
03. INSPECTION BY CIB HARYANA BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ISGEC WORKS, YAMUNA NAGAR (HARYANA)
04. ALL FILLET WELDS ARE 6mm UNLESS OTHERWISE SPECIFIED.
05. FOR DIMENSIONAL TOLERANCE REQUIREMENTS PLEASE REFER DOC. NO. GEN-50990100-PSM-002 AND ITS ASSOCIATED DRAWINGS.
06. CIRCUMFERENTIAL DIMENSION (ON OUTSIDE TRAMMELS ON DRUM) STEAM DRUM - 4875.8 mm.
07. LEGEND :
 - H.W.L. : DENOTES HIGH WATER LEVEL
 - N.W.L. : DENOTES NORMAL WATER LEVEL
 - L.W.L. : DENOTES LOW WATER LEVEL
 - CS-1&CS-5 : DENOTES CLOSING SEAM
 - LS-3&LS-4 : DENOTES LONGITUDINAL SEAM
 - LS-2&CS-3 & CS-4 : DENOTES CIRCUMFERENCE SEAM
 - TL : DENOTES TANGENTIAL LINE
 - S.F.P.W. : SET ON FULL PENETRATION WELD

II) SHOP NOTES :

01. ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUMS.
02. MANHOLE CIRCULAR TYPE SIZE #410 FOR DRUMS REF. DRG. No. JB1161-50010100-006-0001
03. A MAX. 49 mm FILLET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No. : 011-0001 TO 014-0001 AFTER STRESS RELIEVING.
04. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS.
 - i. FOR NOZZLE & STUBS DETAILS REFER DRG.No. : JB1161-50010100-009-0001
 - ii. FOR EXTERNAL AND INTERNAL BAFFLES & PIPE REFER DRG.No. : 010-0001 TO 015-0001.
05. ALL WELDING TO BE DONE BY IBR QUALIFIED WELDER USING IBR WELDING PROCEDURES
06. FOR BOILER DRUMS HAVING SHELL THICKNESS 100mm AND ABOVE FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT:
 - i. ULTRASONIC EXAMINATION OF ALL BUTT WELD BEFORE AND AFTER PWHT.
 - ii. MAGNETIC PARTICLE EXAMINATION OF NOZZLES/STUBS TO SHELL WELDING.
08. ALL THE PADS WELDED ALL-ROUND ON PRESSURE PART SHALL BE PROVIDED WITH 2 Nos. 18" (457mm) TELL-TALE HOLES AT 180° APART ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
09. TOLERANCE FOR FORMED HEADS:-
 - THE INNER SURFACE OF A TORISPHERICAL SHALL NOT DEVIATE OUTSIDE OF THE SPECIFIED SHAPE BY MORE THAN 1/4 % OF NOMINAL INSIDE DIAMETER & INSIDE THE SPECIFIED SHAPE BY MORE THAN 3/4 % OF NOMINAL INSIDE DIAMETER.
10. SUITABLE STIFFENER ARRANGEMENT SHALL BE PROVIDED AT SHOP FOR NOZZLES AT DISHED END TO AVOID DAMAGE DURING TRANSPORTATION.
11. MANHOLE DOORS ARE TO BE PLACED INSIDE THE DRUM BEFORE ASSEMBLING DRUM ENDS FOR FINAL CIRCUMFERENTIAL WELD AS PER ORIENTATION DETAIL.
12. 2mm THK. SS PLATE SHALL BE PROTECTED USING WOODEN BLANKS DURING TRANSPORTATION.
13. PRIOR TO HYDRAULIC TESTING FOLLOWING SHALL BE CARRIED OUT.
 - a) ALL WELD SPATTER, WELD STUBS, SCALE, DIRT ETC SHALL BE REMOVED FROM DRUM.
14. EQUIPMENT SHALL BE THOROUGHLY DRIED/CLEANED AFTER HYDRO TESTING.
15. HYDRO TESTING TO BE CARRIED OUT WITH GASKET THAT IS THE SAME AS THE SPECIFIED SERVICE GASKET.
16. HYDROTEST WATER SHALL NOT CONTAIN MORE THAN 10ppm SUSPENDED SOLIDS.
17. ALL INSIDE EDGES OF NOZZLES AND CONNECTIONS SHALL BE GROUNDED SMOOTH TO 3 MM RADIUS.
18. STRAY ARCS SHOULD BE AVOIDED. IF ANY STRAY ARCS NOTED SHALL BE DRESSED, MPI & UT CHECKED.
19. FOLLOW APPROVED AND APPLICABLE WPS FOR SELECTION OF WELDING CONSUMABLES.
20. WELD SURFACES SHALL BE FREE FROM CRACKS, UNDERCUTS, UNNECESSARY GRINDING, ABRUPT RIDGES AND VALLEYS.
21. VISUAL INSPECTION OF WELD SHALL BE CARRIED OUT AND NOTED BEFORE OFFERING FOR ANY NON DESTRUCTIVE EXAMINATION.
22. HYDRO TEST WATER QUALITY SHALL BE MAINTAINED WITHIN LIMITS SPECIFIED IN DOCUMENT NO.- BLS-PPA-HYD-01-01.

ITEM CODE	ITEM DESCRIPTION	LTYPE	LGROUP	LQTY	LWT (Kg) (TOTAL)	REMARK
1	DETAIL OF DRUM NOZZLE	--	--	1	4923.70	JB1161-50010100-009-0001
2	ADD OF MAN HOLE DOOR	--	--	2	883.38	JB1161-50010100-006-0001
3	CLEAT FOR RING	IS 2062 E250 A	PL.10 THK-50 X 120	8	3.76	-
4	LAGGING RING	IS 2062 E250 A	ISA 75X50X8 THK - 2295 LG.	2	36.26	-
5	SHELL	SA 516 Gr.70	PL.90 THK-2500 X 4960	1	7776.60	(8760.60 Kg.-MAT.FORECAST WT.)
6	SHELL	SA 516 Gr.70	PL.90 THK-2500 X 4960	1	7986.74	(8760.60 Kg.-MAT.FORECAST WT.)
7	SHELL	SA 516 Gr.70	PL.90 THK-1340 X 5110	1	4249.03	(4837.89 Kg.-MAT.FORECAST WT.)
8	SHELL	SA 516 Gr.70	PL.90 THK-2500 X 4960	1	7776.28	(8760.60 Kg.-MAT.FORECAST WT.)
9	DISHED END	SA 516 Gr.70	PL.115 THK-2070 X 2070	1	2818.72	(3868.19 Kg.-MAT.FORECAST WT.)
10	DISHED END	SA 516 Gr.70	PL.115 THK-2070 X 2070	1	2821.99	(3868.19 Kg.-MAT.FORECAST WT.)

SDM001	STEAM DRUM WITH ATTACHMENTS	P	B215	I	39276.04	P.QTY P.WT(Kg) REMARK
P.NO.	PART DESCRIPTION SPECIFICATION	SIZE	LTYPE	LGROUP	LQTY	LWT (Kg) (TOTAL)
ITEM CODE	ITEM DESCRIPTION	LTYPE	LGROUP	LQTY	LWT (Kg) (TOTAL)	REMARK

Issued to	Issued by	Issued date
☐ For pre-planning	☐ For manufacture in shop	
☐ For provisional manufacture in shop	☐ For sub contracting / site work	
This drawing supersedes drawing no. _____ Rev. _____		

W.O. No. PD-0333		/ I.W.T. No : 6662	
CONSULTANT	M/s. AVANT-GARDE ENGINEERS AND CONSULTANTS (P) LTD.		
CLIENT	M/s. RUNGTA MINES LIMITED,CHALIYAMA.		
DO NOT SCALE IF IN DOUBT PLEASE ASK			
THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.			
FOR APPROVAL	NO	FOR INFORMATION	NO
FOR PRODUCTION	YES	FOR ERECTION	YES
RESPONSIBLE DEPT.	DRAWN BY	CHECKED BY	APPROVED BY
MECHANICAL	V.KAVASAKAR	R.G.AMIR/R.KUMARAN	R.ARUNKUMAR
TOTAL WEIGHT IN kg.		PROJECT ID	ELEMENT ID
39276.04		JB1161	50010100
YEAR : 2019		SHEET SIZE : A1	SCALE : 1:31
SERVICE		SINGLE DRUM, WATER TUBE COMPACT CFBC BOILER	
TITLE		ARRANGEMENT & DETAIL OF STEAM DRUM	
DRAWING DOCUMENT ID		JB1161-50010100-007-0001	
REVISION		01	

LIST OF REFERENCE DRAWINGS	
DRAWING DOCUMENT ID	DESCRIPTION
JB1161-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON STEAM DRUM
JB1161-50010100-009-0001	DETAIL OF